MIL-E-80130C
11 September 1984
SUPERSEDING
MIL-E-80130B
26 July 1983

MILITARY SPECIFICATION

ELECTRO-EROSION MACHINE, METAL DISINTEGRATING, ELECTRIC ARC, VIBRATING HEAD, DRILL TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers one type of metal disintegrating machine having either a non-swiveling head or a swiveling head with equipment and accessories specified herein.

1.2 Classification. This specification covers electro-erosion machines of the following classes and sizes as specified (see 6.2.1

Class I

Size	Minimum Table Working Surface Dimensions Inches and Millimeters
1	9" x 15" (23mm x 38mm)
2	15" x 20" (381mm x 508mm)

Class II

Size	Minimum Table Working Surface Dimensions, Inches and Millimeters
1	18" x 20" (457mm x 508mm)
2	25" x 30" (635mm x 762mm)
3	30" x 30" (76.2mm x 76.2mm)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-TES, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3410

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

Class III

Size	- Minimum Table Working Surface Dimensions, Inches and Millimeters
1	30" x 35" (76.2mm x 88.9mm)
2	30" x 50" (76.2mm x 127.0mm)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/NFPA 79 - Metalworking Machine Tools and Plastics Machinery, Electrical Standards for.,

ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951 - Commercial Packaging, Standard Practice for.

ASTM-E-380 - Metric Practice.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First Article. When specified, (see 6.2.1 c), a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.2 Design. The machine shall be new and one of the manufacturer's current models capable of removing broken taps, drills, reamers, screws, in any metal by an electrical erosion process in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1.d). In this specification, all measurements, dimensions, sizes and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in ASTM-E-380.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1.e). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1.1 Dual calibrations. When specified (see 6.2.1.f), measuring and indicating devices shall be graduated in both the US and the SI system of measurements. When a dual US and SI system is furnished, both feed dials shall have independent zero adjustments and both dials shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The machine and all its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1.g).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. The machine shall be designed and constructed such that, under the operating, service, transportation and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state or local statutes in effect at the point of installation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

3.2.9 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings and filler caps shall be accessible.

3.2.10 Interchangeability. To provide for replacement of worn or damaged parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification to the part or the machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. All surfaces to be painted shall be cleaned of all foreign matter. Unless otherwise specified (see 6.2.1.h), the machine shall be painted in accordance with manufacturer's standard commercial practice which shall be not less than one coat of primer and one coat of finish color.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All dials that indicate stock removal or tool movement shall be graduated in increments not larger than 0.10 inch. Dial diameters shall be such that graduations may be read from the normal operating position. Dials shall be permanently and legibly engraved or etched on a non-glare background.

3.3.8 Gears. All gears shall be constructed from a material suitable for the intended purpose. All metal gears shall be machined and heat treated by a process that will impart the hardness, surface finish and toughness that will enable the gear train to transmit full-rated torque of the drive motor without gear damage, failure or premature deterioration and wear. All gears shall be manufactured to definite established standards to provide for interchangeability and parts replacement.

3.3.9 Electromagnetic interference control. When specified (see 6.2.1.i), equipment procured under this specification shall comply with the requirements of MIL-STD-461.

3.4 Components. The components of each machine shall consist essentially of a cabinet enclosed base, worktable, column, coolant system, power supply and disintegrating head, as specified in 3.4.1 through 3.4.5.3.

3.4.1 Base. The base shall be enclosed in a cabinet and shall incorporate, either as an integral part or as a separate unit, provisions for accepting the coolant system and power supply. The base shall be of strength and size to assure maximum rigidity under capacity load (see table I). Swivel type casters, leveling and/or steady rest bolts in the base of the cabinet, and lifting eyes shall be provided.

3.4.2 Worktable. The worktable shall have a machined surface for mounting the workpiece. The worktable shall have not less than the number of T-slots specified in table I. T-slots shall conform to ANSI B5.1. The table shall be provided with a coolant trough extending around the entire periphery of the table work surface and equipped with a coolant drain hose or piping. Positive means of connecting the work piece to the power supply output terminal shall be provided.

3.4.3 Vertical column. The column shall be securely attached to the table or base, and shall support the radial arm and all the components mounted thereon. The column shall be of sufficient length to provide the electrode chuck-to-table surface distance specified in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

3.4.4 Horizontal arm, or ram. The machine shall have either a single horizontal arm, double horizontal arm, folding arm, or ram which supports the head and allows the head to be moved over not less than 70% of the table surface area. The horizontal arm shall be attached to the column in a manner that will allow 360 degrees of arm rotation about the column axis. The arm or ram stroke length shall be such that machining can be performed over the most remote corners of the worktable.

3.4.5 Head assembly. The head assembly for all machines shall have not less than the following motion for the vibrating head:

a. Vertically within the quill housing.
b. Vertically up and down the superstructure column.
c. Horizontally about the superstructure column (360 degree rotation).
d. Horizontally in and out from the superstructure column, and shall permit locating the vibrating head over not less than 75% of the worktable.

Means shall be provided to lock the head (and related components) in any desired position within its horizontal and vertical adjustment range. Locking means shall provide quick and secure clamping without scoring or distorting the mating sliding parts. The vertical position of the head assembly shall provide the electrode-chuck to table-surface distance as specified in table I.

3.4.5.1 Class I. The class I machine shall have a quill housing, a quill and a vibrating head which is removable from the quill and mountable in a different machine by means of a 1/2" diameter arbor located in the top of the head housing. The class I machine shall have a manual and spring controlled gravity feed system. The class I machine shall have a non-swiveling head which holds the electrode axis perpendicular to the table surface, within 2 degrees, in any traversed position of the head.

3.4.5.2 Class II. The class II machine shall be the same as class I, except when specified (see 6.2.1.j), a precision in the quill type vibrating head shall be provided. When specified (see 6.2.1.k), the class II machine shall be provided with an automatic feed system as specified for the class III machine. The head swivel motions of the class II machine shall be the same as class I machines except when specified (see 6.2.1.1), the swivel motion shall be the same as the class III machines. The superstructure crossarm(s) "Ram" shall be capable of being moved up and down the column with a mechanical lift that one operator may operate the full length of the movement.

* 3.4.5.3 Class III. The class III machine shall have a quill housing inside the quill type vibrating head. The class III machine shall have an automatic head feed system consisting of a complete electrical, pneumatic, hydraulic or any combination of the three (in addition to a manual feed system). The automatic feed system shall be such that when the machine is properly adjusted to perform the disintegration process, the cutting electrode will automatically burn into the broken tool, sense any overcurrent,

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

stop or reverse the feed direction, and then continue back into the broken tool after the overcurrent condition has been corrected. The machine shall have an automatic shut-off to prevent overfeeding and to control the depth. Class III machines shall have a head that swivels 360 degrees in a plane perpendicular to the axis of the radial arm or ram. When specified (see 6.2.1.m), the design shall be such that any preset angle between the electrode axis and the table surface is maintained constant, within two degrees, as the head is traversed from any position to any other position. Means shall be provided to clamp the head in any desired position within its swivel range. The superstructure crossarm(s) "Ram" shall be capable of being moved up and down the column with a mechanical lift that one operator may operate the full length of movement.

3.5 Sizes and characteristics. The sizes and characteristics shall be not less than the requirements of table I.

* 3.6 Performance. The class I machines shall be capable of removing 0.040 inch and larger drills and removing size 2-56 thru 1 inch taps. The taps shall be formed of tool steel with characteristics similar to S-4 tool steel with Rockwell hardness of 54C. The class I machine shall also be capable of removing the following size taps within the time limit specified.

Tap Size	Depth	Cutting Time
1/4 - 20	1 in.	1.5 minutes
1/2 - 13	1 in.	5.0 minutes
3/4 - 10	1 in.	12.0 minutes

The class II machines shall be capable of removing the following size taps within the time limit specified.

Tap Size	Depth	Cutting Time
1/4 - 20	1 in.	1 minute
1/2 - 13	1 in.	4 minutes
3/4 - 10	1 in.	6 minutes
1"	1 in.	15 minutes

The class III machines shall be capable of removing the following size taps within the time limit specified:

Tap Size	Depth	Cutting Time
1/4 - 20	1 in.	1 minutes
3/4 - 10	1 in.	6 minutes
1"	2 in.	15 minutes
1-1/2"	2 in.	25 minutes

The class III machine shall also be capable of cutting a core one inch in diameter ±0.046" not less than 6" long in a period of no more than 90 minutes. The core shall be cut from SAE 1020 steel and shall be accomplished without changing electrodes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

Tap Size	Depth	Cutting Time
1/4 - 20	1 in.	1 minute
1/2 - 13	1 in.	2 minutes
3/4 - 10	1 in.	4 minutes
1"	1 in.	15 minutes

3.7 Electrical equipment. Unless otherwise specified (see 6.2.1), the electrical system shall conform to all requirements in this paragraph. The system shall conform to the requirements of NFPA-79, Electrical Standard for Metalworking Machine Tools. Each machine shall draw all of its electrical power from a single 460 VAC, 3 phase, 60 Hz power source having a fused safety disconnect switch. An identified terminal for grounding the machine when installed shall be mounted in or near the disconnect switch.

TABLE I. Sizes and characteristics.

CHARACTERISTICS	REQUIREMENTS						
	Non Swiveling Head					Swiveling Head	
	Class I		Class II			Class III	
	Size 1	Size 2	Size 1	Size 2	Size 3	Size 1	Size 2
TABLE:							
Table working surface, size in inches	12x14	15x20	18x20	25x30	30x30	30x35	30x50
Number of T-Slots (minimum)	2	2	2	2	2	4	6
Load carrying capacity of Table in lbs.	200	300	500	1000	2000	4000	6000
VIBRATING HEAD:							
Chuck To Table Distance (in inches)	12	16	16	18	22	22	28
Quill travel, within quill housing (in inches)	6	6	9	9	9	16	16
COOLANT SYSTEM:							
Reservoir capacity in gallons (minimum)	10	10	10	10	10	35	35
Pump Capacity (gal/min)	2	2	2	2	2	2	2
Pump Coolant Pressure, (pounds/sq. in.)	60	60	70	70	75	75	75
SUPERSTRUCTURE:							
Column diameter in inches (minimum)	2	2	2	2	2	3	3

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

3.7.1 Motors. Motors shall have ball or roller bearings. Unless otherwise specified (see 6.2.1.o), each motor enclosure shall meet the requirements of a dripproof enclosure.

3.8 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant, per table I. The system shall have means for draining and cleaning and shall include a strainer or baffle which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant.

3.9 Hydraulic system. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Over pressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to insure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level.

3.10 Standard equipment. Unless otherwise specified (see 6.2.1.q), one set of hand tools and six burning electrodes shall be furnished with each machine.

3.11 Optional equipment. Optional equipment shall be furnished as specified (see 6.2.1.r), Typical optional equipment may consist of, but need not be limited to the following:

a. Graphite electrodes for removal of large broken tools, bolts and studs.

b. Magnetic fixtures for holding vibrating head and quill housing over broken tools at remote locations.

c. Low work tables for positioning large parts along side of machine, and to contain the coolant.

d. Special holding fixtures to hold workpieces that are of an irregular shape, to allow for faster setup.

e. Deep hole inspection light for viewing down into deep holes.

f. Drill pulling attachment for pulling drills from parts in place of disintegrating them out.

g. AMP meter for reading the current draw, and knowing the machine is operating at the most efficient settings.

h. Float switch to prevent the machine from being operated at a critical low coolant level.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

i. Chucks to hold all sizes of electrodes to be used on the machine.

j. Means for securing parts on the worktable.

k. T-nut and bolt sets for holding odd shaped parts to the worktable.

l. Hard chrome plated superstructure column and crossarm(s).

m. Slings for lifting machine.

n. Motorized lift for moving the crossarm(s).

o. Remote "on and off" control push buttons at the quill housing for faster operation.

p. Spare vibrating head.

q. Lifting eyes.

3.11.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1.s), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.11.2 Nameplate. Unless otherwise specified (see 6.2.1.t), a nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.12 Technical data. When technical data is required it shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

furnished in accordance with the requirements of the contract.

3.13 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the item to pass the first article tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1) quality conformance inspection shall consist of at least the examination in 4.5 and the tests in 4.6. Failure of the item to pass the examination or tests shall be cause for rejection.

4.5 Examination. The machine and equipment shall be examined to determine compliance with the requirements of 3.2 through 3.11.

4.6 Tests.

4.6.1 Performance test. Each machine shall be prepared for operation and operated in accordance with the manufacturer's handbook of instructions, and the machine shall perform the operations specified in 3.6 for the class and size machine under test. The machine shall be operated with the radial arm extended to its maximum length, and raised to the maximum height on the vertical column. The machine and power supply shall show no signs of overheating, overloading, excessive vibration, or deformation of any part of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

machine while in operation. There shall be no evidence of the electrode welding to the workpiece at any time.

4.6.2 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control testing shall be tested for compliance with 3.3.9.

4.7 Packaging inspection. Packaging shall be inspected to determine compliance with the requirement of section 5.

5. PACKAGING

5.1 Packaging, packing and marking. Unless otherwise specified, packaging, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A packaging, level A or level B packing and special marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The machines covered by this specification are intended to be used for removing broken taps, drills, elongate holes, cut keyways, and cut holes of any shape in any metal.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class and size required, etc. (see 1.2).

c. When first article is required for inspection and approval (See 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Dual calibrations (US and SI), if required (see 3.2.1.1.1).

g. Additional safety and health requirements, if required (see 3.2.5).

h. Painting, if different (see 3.3.5).

i. Electromagnetic interference control, if required (see 3.3.9).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

j. A precision in the quill type vibrating head, if required (see 3.4.5.2).

k. When class II, size 2 or 3 machines with an automatic feed system is required (see 3.4.5.2).

l. When the head of class II machine shall swivel the same as class I machines (see 3.4.5.2).

m. When the preset angle between the electrode axis and the table surface is to be maintained constant as head is traversed to any other position (see 3.4.5.3).

n. Electrical system and input power supply if different (see 3.7).

o. Motor enclosure, if different (see 3.7.1).

q. Sizes and characteristics, if different (see 3.5).

q. Standard equipment, if different (see 3.10).

r. Specify and fully describe optional equipment required; if different (see 3.11).

s. Lubrication chart or plate, if different (see 3.11.1).

t. Nameplate, if different (see 3.11.2).

u. First article inspection, if different (see 4.3).

v. Quality conformance inspection, if different (see 4.4).

w. Level of packaging, packing and marking, if different (See 5.1).

6.2.2 Contract data requirements. All data requirements such as parts lists, wiring diagrams, operating manuals, maintenance manuals, alignment and test results shall be requested on a DD Form 1423, Contract Data Requirements List and shall be furnished in accordance with each specified DD Form 1664, Data Item Description as shown on the DD Form 1423.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, heat, etc. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment) the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. The machine shall be equipped with all points-of-operation guarding normally furnished as standard on the manufacturer's commercial machines supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required, as in most cases, the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1.g.

6.5 Changes from previous issue. The margins of the specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements in this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review Activities:
Army - AL
Navy - SH
Air Force - 84
DLA- GS

User Activities:
Army - ME
Navy - MC
Air Force - 84

Preparing Activity:
DLA-IP

Project Number:
3410-0037

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-80130C	Electro-Erosion Machine, Metal Disintegrating, Electric Arc, Vibrating Head, Drill Type

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.